

MEIXIA LIN

Engineering Systems and Design, Singapore University of Technology and Design
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<https://linmeixia.github.io/>

RESEARCH INTEREST

- Algorithm design for large-scale optimization problems in data science
- Convex regression problems
- Optimization methods for statistical models and graphical models
- Stochastic optimization

EMPLOYMENT

**Engineering Systems and Design,
Singapore University of Technology and Design**

Jun. 2022 - Present

- Assistant Professor

**Institute of Operations Research and Analytics,
National University of Singapore**

Jun. 2021 - May. 2022

- Research Fellow
- Mentor: Professor Kim-Chuan Toh

**Department of Mathematics,
National University of Singapore**

Nov. 2020 - May. 2021

- Research Fellow
- Mentor: Professor Kim-Chuan Toh

CriAT–Deep Credit Analytical Technologies, Singapore

May. 2019 - Jul. 2019

- Research Intern

EDUCATION

National University of Singapore, Singapore

Aug. 2016 - Oct. 2020

- Ph.D. in Department of Mathematics
- Supervisors: Professor Kim-Chuan Toh, Professor Chao Zhou, Professor Defeng Sun
- Thesis: Efficient Second-order Algorithms for Structured Convex Composite Programming

Nanjing University, China

Sep. 2012 - Jun. 2016

- B.S. in Information and Computing Science

PUBLICATIONS IN REFEREED SCIENTIFIC JOURNALS

[†] equal contribution or alphabetical order; * corresponding author.

1. Chengjing Wang, Peipei Tang*, Wenling He, and **Meixia Lin**, “Learning the hub graphical model with the structured sparsity via an efficient algorithm”, *Journal of Scientific Computing*, 2025, to appear.
2. Yancheng Yuan, **Meixia Lin***, Defeng Sun, and Kim-Chuan Toh, “Adaptive sieving: A dimension reduction technique for sparse optimization problems”, *Mathematical Programming Computation*, 2025. (DOI: <https://doi.org/10.1007/s12532-025-00282-2>)

3. **Meixia Lin**, Ziyang Zeng, and Yangjing Zhang*, “Multiple regression for matrix and vector predictors: Models, theory, algorithms, and beyond”, *Electronic Journal of Statistics*, 2024, Vol. 18(2), pp. 5563–5600.
4. **Meixia Lin**, Defeng Sun, Kim-Chuan Toh, and Chengjing Wang*, “Estimation of sparse Gaussian graphical models with hidden clustering structure”, *Journal of Machine Learning Research*, 2024, Vol. 25(256), pp. 1–36.
5. Zhenning Cai, Bo Lin*, and **Meixia Lin**, “A positive and moment-preserving Fourier spectral method”, *SIAM Journal on Numerical Analysis*, 2024, Vol. 62(1), pp. 273–294.
6. **Meixia Lin**[†], Yancheng Yuan[†], Defeng Sun*, and Kim-Chuan Toh, “A highly efficient algorithm for solving exclusive lasso problems”, *Optimization Methods and Software*, 2024, Vol. 39(3), pp. 489–518.
7. Subhroshekhar Ghosh, **Meixia Lin***, and Dongfang Sun, “Signal analysis via the stochastic geometry of spectrogram level sets”, *IEEE Transactions on Signal Processing*, 2022, Vol. 70, pp. 1104–1117.
8. **Meixia Lin***, Defeng Sun, and Kim-Chuan Toh, “An augmented Lagrangian method with constraint generation for shape-constrained convex regression problems”, *Mathematical Programming Computation*, 2022, Vol. 14, pp. 223–270.
9. **Meixia Lin**, Yong-Jin Liu*, Defeng Sun, and Kim-Chuan Toh, “Efficient sparse semismooth Newton methods for the clustered lasso problem”, *SIAM Journal on Optimization*, 2019, Vol. 29(3), pp. 2026–2052.

PUBLICATIONS IN CONFERENCE PROCEEDINGS

[†] equal contribution or alphabetical order; * corresponding author.

1. **Meixia Lin**, and Yangjing Zhang*, “DNNLasso: Scalable graph learning for matrix-variate data”, International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.
2. Rémi Bardenet[†], Subhroshekhar Ghosh^{†*}, and **Meixia Lin**^{†*}, “Determinantal point processes based on orthogonal polynomials for sampling minibatches in SGD”, Conference on Neural Information Processing Systems (NeurIPS), 2021. (**Spotlight** presentation, less than 3% acceptance rate)

PREPRINTS

[†] equal contribution or alphabetical order; * corresponding author.

1. Lei Yang, Liwei Luo, and **Meixia Lin***, “Shadow-point enhanced inexact accelerated proximal gradient method with preserved convergence guarantees”, submitted, April 2025.
2. **Meixia Lin**, and Yangjing Zhang*, “Low rank convex clustering for matrix-valued observations”, submitted, December 2024.
3. Hong T. M. Chu, **Meixia Lin***, and Kim-Chuan Toh, “Wasserstein distributionally robust optimization and its tractable regularization formulations”, submitted, May 2024.

RESEARCH GRANTS

- Principal Investigator, MOE Academic Research Fund Tier 2 (2024-2027), SGD \$680k.
- Principal Investigator, TL Seed Research Project Grant (2023-2025), SGD \$96k.
- Principal Investigator, SUTD Kickstarter Initiative (2022-2025), SGD \$477k.

TEACHING EXPERIENCES

Singapore University of Technology and Design

- 42.515 Machine Learning and Analytics

Spring 2025

- 40.319 Statistical and Machine Learning *Spring 2023, Spring 2025*
- 10.022 Modelling Uncertainty *Autumn 2022, Autumn 2023, Autumn 2024*
- 40.016 The Analytics Edge *Summer 2022*

National University of Singapore

- Teaching Assistant *Aug. 2017 - May. 2020*

GRADUATE STUDENT SUPERVISION

Graduate Students

- Wong Wei Jin Justin, STEP Master's student at SUTD, November 2024 –
- Mr Zhihao Liu, PhD candidate at SUTD, September 2023 –

Postdocs and Research Assistants

- Dr Yuexin Zhou, Postdoc, April 2025 –
- Mr Qirui Yang, Research Assistant, February 2025 –
- Dr Wenhao Lu, Postdoc, October 2024 –
- Dr. Bonan Hou, Research Assistant, October 2024 –
- Dr Xiaoyu Dong, Postdoc, May 2024 – November 2024
Current position: Lecturer, School of Mathematics, Statistics and Mechanics, Beijing University of Technology, China
- Dr Qian Zhang, Postdoc, September 2023 –
- Mr Tirtho Sarker, Research Assistant, March 2024 – February 2025

Visiting Scholars

- Miss Mengjiao Shi, Visiting PhD Student from Henan University, August 2024 – March 2025
- Associate Professor Sisi Zheng, Visiting Scholar from Huizhou University under the support of the Chinese Scholar Council (CSC), March 2024 – January 2025
- Mr Jiawei Gu, Visiting PhD Student from Nanjing University of Science and Technology, October 2023 – April 2024
- Mr Ziyang Zeng, Visiting Master Student from National University of Singapore, September 2023 – September 2024
- Miss Chu Thi Mai Hong, Visiting PhD Student from National University of Singapore, March 2023 – April 2023

CONSULTING PROJECT

Tokopedia

Jan. 2022 - Oct. 2023

PROFESSIONAL ACTIVITIES

Minisymposium Organizer in International Conferences

- Chair, Minisymposium on “Recent Advances in Structured Non-Smooth Optimization” (virtual), SIAM Conference on Optimization 2021 (OP21), Jul. 20–23, 2021.

Invited Talks

- “*Sampling minibatches in SGD – A DPP based approach via Orthogonal Polynomials*”, 10th International Congress on Industrial and Applied Mathematics, Tokyo, Japan, Aug. 20–25, 2023.
- “*Sampling minibatches in SGD – A DPP based approach via Orthogonal Polynomials*”, Workshop on Optimization in the Big Data Era, Singapore, Dec. 05–16, 2022.
- “*Signal analysis via the stochastic geometry of spectrogram level sets*”, Workshop on Determinantal and Permanent Point Processes, Quantum Physics, and Signal Processing, Lyon, France, May. 30–Jun. 10, 2022.
- “*Estimation of sparse Gaussian graphical models with hidden clustering structure*” (virtual), SIAM Conference on Optimization 2021 (OP21), Jul. 20–23, 2021.
- “*Adaptive sieving with PPDNA: Generating solution paths of exclusive lasso models*” (virtual), Forum on Operations Research and Information 2020, Beijing, China, Nov. 23–26, 2020.
- “*Efficient sparse semismooth Newton methods for the clustered lasso problem*”, Workshop on Matrix Optimization 2019, Beijing, China, Nov. 29–30, 2019.
- “*A dual Newton based preconditioned proximal point algorithm for exclusive lasso models*”, The Sixth International Conference on Continuous Optimization, Berlin, Germany, Aug. 3–8, 2019.
- “*Efficient sparse semismooth Newton methods for the clustered lasso problem*”, International Symposium on Mathematical Programming 2018, Bordeaux, France, Jul. 1–6, 2018.

SERVICES

Committee Services (in SUTD)

- **Fellow of SUTD Academy**
- Member of SUTD Institutional Review Board
- Undergraduate (UG) Committee (in ESD)
- Research And Scholarly Excellence (RASE) Committee (in ESD)
- Focus Track Lead of Business Analytics and Operations Research (in ESD)

University Services

- **Vice president** of NUS SIAM Student Chapter, Jul. 2019 – Jun. 2020.

HONORS AND AWARDS

- **Best Graduate Researcher Award** in Department of Mathematics, NUS, Singapore, 2020.
- Outstanding Graduate Award, Nanjing University, China, 2016.
- First-class Scholarship of Top-notch Student Training Plan, China, 2015.
- First-class People’s Scholarship, China, 2014.
- **National Scholarship**, China, 2013.